

Computational Fluid Dynamics

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We want to simulate the flow of fluid with kinematic viscosity $\nu = 0.1$ through the bifurcating duct sketched in Figure 1.

The problem is governed by the steady incompressible Navier-Stokes equations, with Dirichlet boundary conditions imposed on the inflow left boundary

$$\mathbf{u} = [(2 - y)(2 + y)/4, 0]^T, \quad \text{on } \Gamma_1,$$

and on the duct walls

$$\mathbf{u} = [0, 0]^T, \quad \text{on } \Gamma_4,$$

(as indicated in the Figure). The outflow boundary of the short upper branch is Γ_3 , while the longer lower branch is numerically truncated at Γ_2 . The problem is therefore solved in the shaded domain Ω .



Figure 1: Computational domain

- Introduce the strong formulation of the steady incompressible Navier-Stokes equations for the solution of the problem and complete the setting by selecting suitable boundary conditions on Γ_2 and Γ_3 .
- Introduce the Newton linearization of the problem and the weak formulation of the linear problem to be solved at each Newton iteration.
- Based on the template `template080224.ipynb`, write a **Firedrake** notebook for the solution of problem using $\mathbb{P}_2 - \mathbb{P}_1$ finite elements for velocity and pressure, respectively, starting from the Stokes solution and defining a suitable convergence criterion for the Newton method.
- Save the plots of the velocity and pressure fields obtained at the previous point (export them as files `SURNAME_NAME_velocity.png`, `SURNAME_NAME_pressure.png`) and compute the flowrate through the two outflow boundaries Γ_2 and Γ_3 , as well as the L^2 -norm of the vertical velocity u_y on Γ_2 and Γ_3 . Comment the obtained results.
- Modify the solver considering now a fixed-point Picard iteration for the solution of the nonlinear problem and compare its performance with the one of the Newton method.